- How quickly the IT team is notified when something is wrong or beginning to fail
- How rapidly the IT team can triage the problem, understand its impact and effectively provide any mitigation
- Effectiveness of the IT team in finding the underlying cause and fixing it

Observability includes contextual telemetry, dynamic service topology discovery, anomaly detection, proactive issue avoidance, alert grouping, correlation, root cause identification, advice on next best action, visualisation of the severity of business impact, and cost optimisation.

Observability patterns for cloud workspaces

Enterprises are adopting cloud services to take advantage of the benefits they promise. Observability frameworks help to measure the actual business value (in terms of reliability, cost, quality of service, etc) enterprises achieve from cloud adoption. The cloud workspace observability framework should be designed keeping in mind an enterprise's business goals. The primary patterns that the cloud workspace observability framework should deploy are listed below.

Service reliability and health monitoring patterns: Insights into the health of workspaces is important. Availability, responsiveness, user experience measures are major items to be monitored.

Service reliability engineering (SRE) golden signals define what it means for the service to be healthy. Workspace service observability should monitor below golden signals in the context of desktop service.

- *Latency:* This is the time taken to serve a request. The duration a service takes to make the desktop ready to use from the time of the desktop request event is provision latency. Round trip time (RTT) is the network latency.

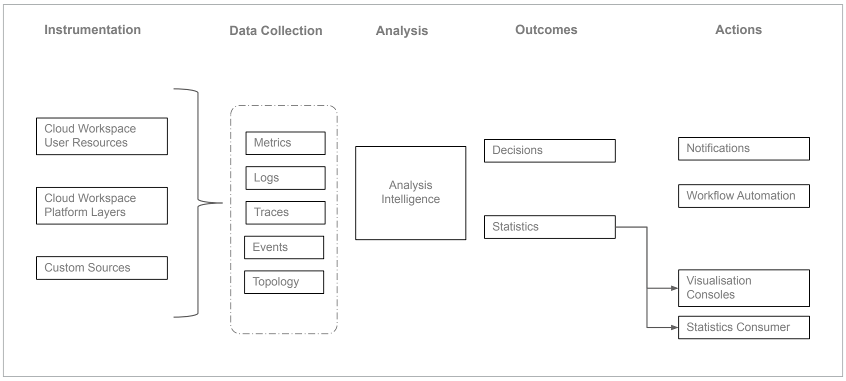

Figure 1: Cloud workspace observability blueprint

- *Traffic:* This is the stress due to the demand on the system. This signal is critical for desktop brokering, session gateway and other service mediation services.
- *Errors:* This is the rate of requests that are failing. Connection failures fall under this category.
- *Saturation:* This is the overall capacity of the system. This signal becomes critical for multi-session desktops, where a single virtual machine is shared among many users. Host server CPU, RAM utilisation, and queue depth fall under this category.

Security and governance patterns: Data security is a prime concern for enterprises, especially in remote working. Suspicious behaviour such as unusual access times, unusual locations, numerous logon failures, unusual file downloads, excessive authentication failures, abnormal permission changes are some sample threat signals that need to be monitored in real-time and acted upon immediately. Well-architected frameworks implement security patterns to combat such attacks and improve the cloud security posture. Patterns such as zero trust, role based permissions, conditional and contextual access, multi-factor authentication, multi-layer granular controls are some of the important ones that workspaces should adopt. The framework should ensure that these patterns are applied properly. To ensure real-time traffic monitoring and threat detections, observability frameworks should adopt AI/ML algorithms. These patterns have been quite visible in recent times with cloud native security information and event management (SIEM) solutions.

Cost optimisation patterns: Enterprises migrate to the cloud for leveraging opex and flexible commercial models. But if there is no judicious control over the cloud resource usage, the result could be frustrating. Cloud commercials are based on actual resource usage. It is the IT team's responsibility to monitor resource usage and shutdown or release unnecessary resources. Choosing an optimal billing model is also very important. An observability framework should monitor resource usage and provide guidance on the optimal approach. It should implement cost optimisation patterns such as auto scaling, optimal charge model switching, auto stopping, etc.

Automation patterns: Observability patterns demand real-time discovery and automated orchestration of analysis and healing logic. They help in immediate reaction, and avoid human errors and delays. To help enterprises in achieving these goals, all public cloud vendors recommend well-architected cloud native automation patterns for best results. For example, serverless functions are best suitable for deploying automation patterns.